Bridging the Tap:

A Data-Driven Assessment of SDG 6 Progress Using WHO/UNICEF JMP Data

📌 Project Architecture (STAR Framework)

Executive Summary

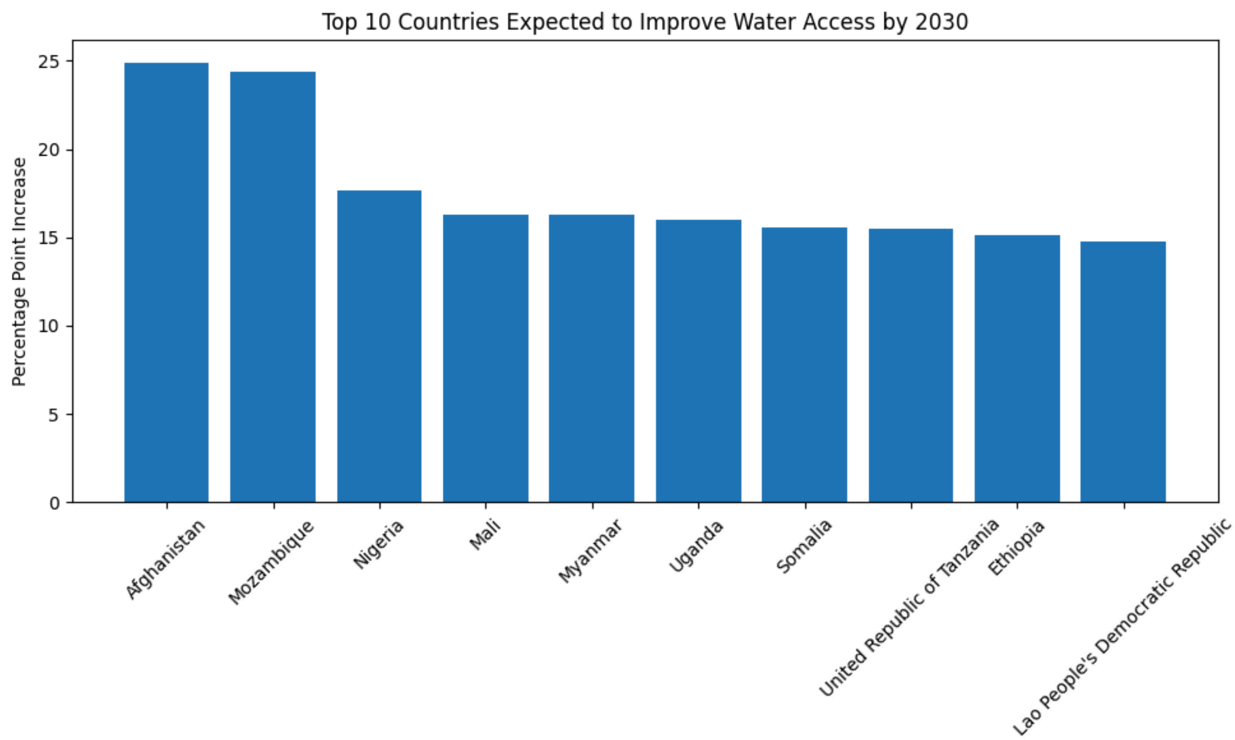
- **Situation:** Tracked global progress toward United Nations Sustainable Development Goal 6 (Universal Access to Clean Water and Sanitation) leveraging raw tracking data from the WHO/UNICEF Joint Monitoring Programme (JMP).
- **Task:** Overcame severe temporal reporting fragmentation across over 200 global entities to calculate true historical progress velocities and mathematically forecast national outcomes for the 2030 UN deadline.
- **Action:** Engineered a dynamic algorithmic pipeline in Python (Pandas & Scikit-Learn) using custom date-boundary filters to bypass chronological data gaps. Stratified results into non-linear machine learning models and structural risk categories (*On Track, Near Target, Moderate Risk, High Risk*).
- **Result:** Deployed a fully responsive, 4-page interactive dashboard via v0 (Vercel) featuring global risk heatmaps, regional variance matrices, and model validation ledgers designed to optimize strategic capital deployment for NGOs.

🔮 The "SDG 2030 Deadline" Velocity Analysis

Core Analytical Objective

Traditional development reporting often highlights where a country stands today, but fails to account for momentum. This study addresses a critical operational question for international stakeholders:

> *“Based on historical progress velocities from 2000 to the present, which countries and regions are statistically projected to achieve the UN SDG 6 target of 100% universal water access by 2030, and where must structural interventions be deployed immediately?”*



Methodology & Mathematical Framework

To calculate individual national trajectories without distorting historical boundaries, the pipeline evaluates metrics through a three-stage mathematical engine:

1. **Historical Annual Progress Rate (Velocity):** Evaluates the average annualized percentage point change between a country's earliest available baseline (**V**start) and its latest recorded data point (**V**end) across the available time horizon (**T**).
   $Annual\,Velocity\ = Vend\ -\ Vstart\ \div\ Tend\ -\ Tstart$
2. **The 2030 Target Projection:** Extrapolates current momentum forward from the latest observed reporting year to the UN deadline of 2030.
   $Projected\ 2030\ Coverage\ = Vend\ +\ (Annual\,Velocity\ \times\ (2030\ -\ Tend))$
3. **The SDG Target Gap:** Quantifies the precise percentage point deficit remaining to reach absolute universal access.
   $SDG\ 2030\ Gap\ =\ 100\%\ -\ Projected\ 2030\ Coverage$

Overcoming Data Engineering Roadblocks

💡 **Data Resilience Note:** While compiling the initial tracking models, applying strict static chronological filters resulted in severe data drops due to country-specific reporting intervals. To prevent sample size degradation, the pipeline was refactored to dynamically establish custom mathematical boundaries for each country using index optimization (idxmin() and idxmax()). This process scaled data integrity from an initial 18-country cohort to a comprehensive global evaluation matrix spanning 213 records.

🎯 Strategic Diagnostics & Resource Allocation

By comparing linear baseline trajectories against machine learning curve forecasts, the interactive dashboard uncovers specific operational bottlenecks requiring targeted funding shifts:

🏝️ 1. The Caribbean Gridlock (High-Baseline Stagnation)

- **The Data Reality:** A dense cluster of territories—including Anguilla, Antigua and Barbuda, Aruba, the Bahamas, and the Cayman Islands—exhibit an identical Annual_Velocity coefficient of exactly 0.000000.
- **The Diagnostic Insight:** Despite maintaining high static historical baselines ($\ge 85\%$), infrastructure development in these zones has reached an absolute standstill. This stagnation indicates that rising climate vulnerability, intense seasonal weather events, and over-indexing on tourism-centric municipal grids are locking access rates in place.
- **Actionable Policy Decision:** International donors must immediately shift funding away from standard expansion projects in the Latin America & Caribbean region and pivot capital toward **climate-resilient, decentralized water storage technologies**.

🌍 2. The Momentum Leaders (High-Impact Acceleration)

- **The Data Reality:** Countries such as Equatorial Guinea, Comoros, Somalia, and Uganda are flagged with significant positive baseline changes (e.g., Somalia and Uganda accelerating at $>1.5\%$ per year), yet they remain classified as "High Risk" or "Off Track" due to their low historical starting points.
- **The Diagnostic Insight:** These nations represent operational successes, not development failures. Their underlying water infrastructure programs possess highly effective deployment frameworks, but they lack the macro-scale funding necessary to outpace the 2030 deadline.
- **Actionable Policy Decision:** These regions offer the highest return on human impact per dollar spent. Rather than designing new, unproven aid programs, NGOs should **inject scaling capital directly into these active networks** to compress timeframes and move these nations into the "On Track" quadrant before 2030.